Project Design Phase-II

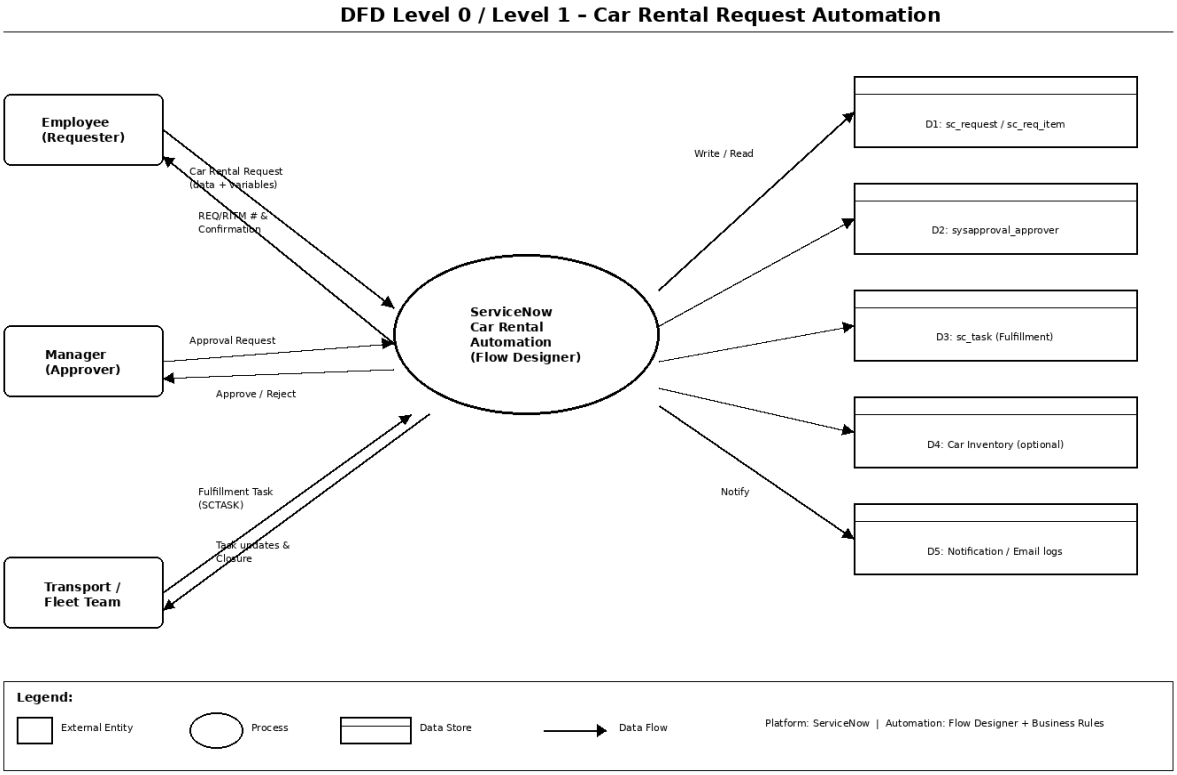
Data Flow Diagrams & User Stories

Date	Team ID	Project Name	Maximum Marks
11 october 2026		Car Rental Request Automation	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) is a visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 / Level 1 – Car Rental Request Automation



External Entities & Major Data Flows:

- Employee (Requester) → submits Car Rental Request data (dates, locations, car type, reason)
- Manager → receives Approval Request; returns Approve / Reject decision
- Transport / Fleet Team → receives Fulfillment Task; updates car assignment & task status
- ServiceNow Platform (Central Process) → creates REQ / RITM / SCTASK / Approval records; sends notifications
- Notification Engine → email / portal messages to all parties

Core Data Stores:

- D1: sc_request / sc_req_item (REQ & RITM) – request header and item details
- D2: sysapproval_approver – approval records and decisions
- D3: sc_task – fulfillment tasks for Transport team
- D4: Custom Car Inventory table (optional) – available vehicles, status, assignment history
- D5: sys_email / notification logs – communication history

DFD Level 1 – Major Processes

1. P1 – Submit Catalog Request: Employee fills variables → creates REQ + RITM
2. P2 – Route for Approval: Flow Designer creates Approval record → Manager decision
3. P3 – Create Fulfillment Task: On approval → SCTASK assigned to Transport group
4. P4 – Fulfill & Close: Transport updates task → RITM / REQ closed; notifications sent
5. P5 – Notify Stakeholders: Triggered at each state change (submission, approval, rejection, closure)

User Stories

User stories for the Car Rental Request Automation product.

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Priority	Release
Employee	Catalog Request	USN-1	As an employee, I can open the Service Catalog item “Car Rental Request” and submit a request with date, locations, duration, car type and reason.	REQ & RITM created; confirmation number shown.	High	Sprint-1
		USN-2	As an employee, I receive an email / portal notification confirming my car rental request submission.	Notification received with request number and summary.	High	Sprint-1
	Status Tracking	USN-3	As an employee, I can view the current status of my car rental request from My Requests / Service Portal.	Status and activity stream visible and up-to-date.	High	Sprint-1
		USN-4	As an employee, I receive a notification when my request is approved, rejected or completed.	Correct notification with reason (if rejected) or next steps.	High	Sprint-1
Manager	Approval	USN-5	As a manager, I receive a notification and can approve or reject a car rental request assigned to me with optional comments.	Approval record updated; request advances or is cancelled.	High	Sprint-1
		USN-6	As a manager, I can see key request details (dates, locations, car type, reason) on the approval form.	All relevant variables displayed on approval UI.	Medium	Sprint-1
Transport Team	Fulfillment	USN-7	As a Transport team member, I receive an automatic SCTASK when a car rental request is approved.	Task appears in group queue with correct assignment and variables.	High	Sprint-1
		USN-8	As a Transport team member, I can update the task (assign car, add work notes) and close it, which closes the parent request.	Task closure updates RITM/REQ state; requester notified.	High	Sprint-1
Admin	Config & Reporting	USN-9	As an admin, I can capture all configuration changes in an Update Set for promotion.	Update Set contains all relevant records and can be committed on target instance.	High	Sprint-1
		USN-10	As an admin, I can view reports / dashboards showing open car rental requests, cycle times and SLA performance.	Reports filterable by department, car type, status; data accurate.	Medium	Sprint-2

Note: All user stories map to Flow Designer stages, Business Rules and Catalog configuration. Implementation follows ServiceNow best practices (Update Sets, ACLs, Flow error handling).